CaseGraph build order and model map

50 features across four judiciary products, in the order to build them. Work one row at a time, top to bottom.

Basic ML · 9 | Deep model · 9 | Vision · 3 | GenAI · 8 | Graph · 6 | Engineering · 15

Part 1. What to build, in order

How to read this: the gold **Order** column is your to-do list. Start at 1 and do not skip ahead; each milestone only needs what came before it. **#** is the feature's fixed id, used in Part 2 and in the code. Six milestones: foundation, pendency models, legal research, bail with vision, consumer complaints, platform hardening.

ORDER	#	FEATURE	MODEL OR TECHNIQUE NEEDED	TYPE	PHASE
Milestone 1 - Foundation: data in, graph up, guardrails on No ML yet. Get real court data flowing into Neo4j and build the two safety pieces every later feature depends on.					
1	1	**eCourts and NJDG case status ingestion by CNR, scheduled refresh**	Scraper + scheduler (no model)	Engineering	1
2	2	**Judgment ingestion from Indian Kanoon, SC and HC sites with metadata**	Scraper + regex/NER metadata parser	Engineering	1
3	4	**Statute graph: IPC/BNS, CrPC/BNSS, Evidence/BSA, CPA 2019 with old-to-new mapping**	Rule-based section parser + Neo4j schema + mapping table	Graph	1
4	14	**Pendency dashboard by state, district, court, case type**	Aggregation queries (no model)	Engineering	1
5	17	**Matter tracker with alerts on new orders**	Scheduled diff + notifications (no model)	Engineering	1
6	7	**Hybrid retrieval: vector embeddings + Cypher**	Sentence embedding model (bge-m3 or e5) + vector index + Cypher templates	Deep model	1
7	25	**Citation verifier: every cited case must exist in graph**	Deterministic regex + graph lookup (no model)	Graph	1
8	29	**Old-to-new code cross reference in search and drafts**	Mapping table + Cypher (no model)	Graph	1
9	39	**Mandatory human review gate before draft release**	LangGraph interrupt / approval node (no model)	Engineering	1
Milestone 2 - First models: pendency predictor Your first basic ML, first explanation with an LLM, and first deep model, all on the same hearing data.					
10	9	**Time-to-disposal prediction with confidence range**	Gradient boosting regression (LightGBM) or survival model (Cox / random survival forest)	Basic ML	1
11	11	**Adjournment likelihood for next hearing**	Binary classifier (logistic regression baseline, then GBM)	Basic ML	1
12	13	**Benchmark against similar cases in same court**	kNN on case feature vectors + Cypher filters	Basic ML	2
13	15	**Hearing gap anomaly detection**	Isolation forest or z-score on inter-hearing gaps	Basic ML	2

ORDER	#	FEATURE	MODEL OR TECHNIQUE NEEDED	TYPE	PHASE
14	12	Delay driver explanation in plain language	SHAP feature attributions + LLM narrative	GenAI	2
15	10	Stage-wise stall prediction	Multiclass GBM, or sequence model (LSTM / Transformer) over hearing history	Deep model	2
Milestone 3 - Legal research: embeddings and citation graph Deep models and GenAI on judgment text. The citation graph becomes useful here.					
16	19	Brief-to-precedent semantic search	Embedding retriever + cross-encoder reranker	Deep model	1
17	26	Judgment summariser: facts, issues, holding, ratio	LLM structured extraction; optional fine-tuned seq2seq (Flan-T5 / mT5)	GenAI	2
18	5	Citation graph with typed edges: cites, follows, distinguishes, overrules, interprets	Citation regex + relation classifier (fine-tuned LegalBERT / InLegalBERT)	Deep model	2
19	21	Treatment status: followed, distinguished, overruled	Relation classifier on citing sentences (fine-tuned LegalBERT)	Deep model	2
20	20	Interactive citation graph view for any judgment	Neo4j traversal + graph visualisation (no model)	Graph	2
21	23	Statute-centric timeline of interpreting judgments	Cypher query over interprets edges (no model)	Graph	2
22	24	Research memo drafting with verified citations only	LLM in LangGraph with retrieval and verifier node	GenAI	2
23	6	Entity resolution for judges, courts, advocates, parties	String similarity features + logistic regression or XGBoost match classifier	Basic ML	2
24	8	Document deduplication and versioning	MinHash / SimHash near-duplicate detection	Basic ML	2
Milestone 4 - Bail and undertrial: vision enters Rule engine first, then the vision model reads scanned remand orders, then drafting behind the review gate.					
25	31	Eligibility computation under BNSS Section 479	Rule engine over statute graph (no model)	Engineering	1
26	33	Offence severity and maximum sentence lookup	Statute graph query (no model)	Graph	1
27	3	Scanned document OCR for orders, FIRs, charge sheets, remand slips	OCR (PaddleOCR or Tesseract) + vision LLM for layout and fields	Vision	2
28	32	Custody period from remand and production orders	Vision LLM or OCR + date/entity extraction	Vision	2
29	35	Bail precedent finder for similar offence and facts	Embedding similarity on fact patterns + filters	Deep model	2
30	34	Bail application draft generator, court-specific format	LLM with templates in LangGraph	GenAI	2
31	37	Prison-wise and district-wise undertrial dashboard	Aggregation queries (no model)	Engineering	2
Milestone 5 - Consumer complaints Reuse vision and drafting from Milestone 4 on a simpler, high-volume workflow.					

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32	41	Forum and jurisdiction finder by claim value and location	Rule engine on pecuniary limits (no model)	Engineering	1
33	42	Limitation period check with reminders	Rule engine + date extraction (no model)	Engineering	1
34	40	Evidence intake from photos of bills, warranty cards, chat screenshots	Document type classifier (small CNN / ViT) + vision LLM field extraction	Vision	2
35	43	Complaint drafting in prescribed format with relief computation	LLM with templates	GenAI	2
36	44	Similar order retrieval from NCDRC and State Commissions	Embedding retriever over orders	Deep model	2
37	45	Outcome and compensation estimate from past orders	Outcome classifier + compensation regression (GBM)	Basic ML	3
Milestone 6 - Platform hardening and later features Multi-user SaaS plumbing, retraining loop, then every Phase 3 feature.					
38	49	Role-based access: litigant, lawyer, legal aid, researcher	Application feature (no model)	Engineering	2
39	50	Agent audit trail and feedback loop to retrain classifiers	LangGraph tracing (LangSmith) + retraining pipeline (MLflow)	Engineering	2
40	16	Next hearing outcome: adjourn, progress, dispose	Sequence model (GRU / small Transformer) on hearing event sequence	Deep model	3
41	22	Conflicting bench detection across High Courts	NLI / contradiction model on holdings + graph queries	Deep model	3
42	27	Headnote and issue generation	LLM generation with few-shot examples	GenAI	3
43	28	Argument builder: authorities for and against	LLM + retrieval + stance classifier	GenAI	3
44	36	Legal aid queue prioritised by eligibility and custody time	Ranking model (GBM) or weighted score	Basic ML	3
45	18	Alternative route suggestion: Lok Adalat, mediation, fast-track	Rule engine + eligibility classifier	Basic ML	3
46	38	Hearing reminder and bail condition compliance tracker	Scheduler + notifications (no model)	Engineering	3
47	46	e-Daakhil filing checklist and fee calculator	Rule engine (no model)	Engineering	3
48	47	Post-filing case timeline tracker	Scheduler + eCourts sync (no model)	Engineering	3
49	30	Saved research workspaces with DOCX and PDF export	Application feature (no model)	Engineering	3
50	48	Multilingual question answering and drafting (Hindi, regional)	Multilingual LLM + IndicTrans2 translation model	GenAI	3

Part 2. The same 50 features grouped by product

> Reference view. The **Order** column repeats the build position from Part 1 so you can see where each product's features fall in the sequence.

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A. Data ingestion and knowledge graph					
1	1	**eCourts and NJDG case status ingestion by CNR, scheduled refresh**	Scraper + scheduler (no model)	Engineering	1
2	2	**Judgment ingestion from Indian Kanoon, SC and HC sites with metadata**	Scraper + regex/NER metadata parser	Engineering	1
27	3	**Scanned document OCR for orders, FIRs, charge sheets, remand slips**	OCR (PaddleOCR or Tesseract) + vision LLM for layout and fields	Vision	2
3	4	**Statute graph: IPC/BNS, CrPC/BNSS, Evidence/BSA, CPA 2019 with old-to-new mapping**	Rule-based section parser + Neo4j schema + mapping table	Graph	1
18	5	**Citation graph with typed edges: cites, follows, distinguishes, overrules, interprets**	Citation regex + relation classifier (fine-tuned LegalBERT / InLegalBERT)	Deep model	2
23	6	**Entity resolution for judges, courts, advocates, parties**	String similarity features + logistic regression or XGBoost match classifier	Basic ML	2
6	7	**Hybrid retrieval: vector embeddings + Cypher**	Sentence embedding model (bge-m3 or e5) + vector index + Cypher templates	Deep model	1
24	8	**Document deduplication and versioning**	MinHash / SimHash near-duplicate detection	Basic ML	2
B. Case pendency and delay predictor					
10	9	**Time-to-disposal prediction with confidence range**	Gradient boosting regression (LightGBM) or survival model (Cox / random survival forest)	Basic ML	1
15	10	**Stage-wise stall prediction**	Multiclass GBM, or sequence model (LSTM / Transformer) over hearing history	Deep model	2
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C. Legal research assistant					
16	19	Brief-to-precedent semantic search	Embedding retriever + cross-encoder reranker	Deep model	1
20	20	Interactive citation graph view for any judgment	Neo4j traversal + graph visualisation (no model)	Graph	2
19	21	Treatment status: followed, distinguished, overruled	Relation classifier on citing sentences (fine-tuned LegalBERT)	Deep model	2
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D. Bail and undertrial monitoring					
25	31	Eligibility computation under BNSS Section 479	Rule engine over statute graph (no model)	Engineering	1
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E. Consumer court complaint helper					
34	40	**Evidence intake from photos of bills, warranty cards, chat screenshots**	Document type classifier (small CNN / ViT) + vision LLM field extraction	Vision	2
32	41	**Forum and jurisdiction finder by claim value and location**	Rule engine on pecuniary limits (no model)	Engineering	1
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47	46	**e-Daakhil filing checklist and fee calculator**	Rule engine (no model)	Engineering	3
48	47	**Post-filing case timeline tracker**	Scheduler + eCourts sync (no model)	Engineering	3
F. Cross-cutting					
50	48	**Multilingual question answering and drafting (Hindi, regional)**	Multilingual LLM + IndicTrans2 translation model	GenAI	3
38	49	**Role-based access: litigant, lawyer, legal aid, researcher**	Application feature (no model)	Engineering	2
39	50	**Agent audit trail and feedback loop to retrain classifiers**	LangGraph tracing (LangSmith) + retraining pipeline (MLflow)	Engineering	2

Sources: eCourts, NJDG, Indian Kanoon, SC/HC websites, NCDRC orders. Stack: Neo4j, LangChain, LangGraph, scikit-learn/LightGBM, PyTorch, vision LLM.

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